Belief-Constrained Inference

Correcting Visual Premises in VLMs at Inference Time

CV 2026

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github.com/ksananth4424/BCI_Inference_Time

Outline

1. Motivation & Problem
2. Method: Belief-Constrained Inference (BCI)
3. Experimental Setup
4. Results
5. Analysis
6. Conclusion & Future Work

VLMs Can Reason Correctly but Perceive Wrongly

The Core Challenge

When a VLM answers incorrectly, **why** did it fail?

- **Reasoning Error (RE):** Faulty logic on mostly correct perceptual beliefs
- **Premise Error (PE):** Correct logic on *false* beliefs about the scene

Standard evaluation collapses both failure modes; no distinction, no targeted fix.

Motivating Example

Image: A red cup on a wooden table

VLM Belief: "the cup is blue"
VLM Answer: "The blue cup is on the table"

Reasoning: ✓ OK **Premise: ✗ WRONG**

The Epistemic Failure Hypothesis

Central Question

How often are VLM errors caused by false visual premises, and can correcting those premises at inference time recover the correct answer **without** *updating model weights?*

Hypothesis

A substantial fraction of VLM failures are **epistemic**:

- Wrong answer $\Leftarrow$ wrong scene representation
- Not because the reasoning chain is flawed

$\Rightarrow$ Selectively correcting false beliefs should flip wrong answers to correct.

Causal Claim

Under a **random-ablation** control (removing beliefs without targeted correction), we should *not* observe the same gains.

This distinguishes *targeted* premise correction from a generic perturbation effect.

Why Existing Approaches Fall Short

Training-Time & Post-Hoc

- RLHF, regularisation: require weight updates, cannot be applied post-deployment
- Woodpecker: corrects beliefs *after* answer generation; does not close the reasoning loop at premise level

Output-Level Inference (VCD, Self-Consistency)

- Operate at *output distribution* level
- No interpretability, no error attribution

BCI's Approach

Intervene at **intermediate belief level** i.e. before the final answer for interpretable, targeted, and causally validated correction.

Key Differentiator

BCI quantifies the **causal contribution** of premise errors via controlled random ablation; no prior method does this.

Problem Formulation

Formal Setup

Let (I, Q, A^*) denote an image, question, and ground-truth answer. A VLM $\mathcal{M}$ produces $\hat{A} = \mathcal{M}(I, Q)$.

Premise Error (PE)

$\hat{A} \neq A^*$ **and** at least one extracted belief $b_i \in \mathcal{B}$ is verifiably false w.r.t. I.

Root cause: wrong scene representation

Reasoning Error (RE)

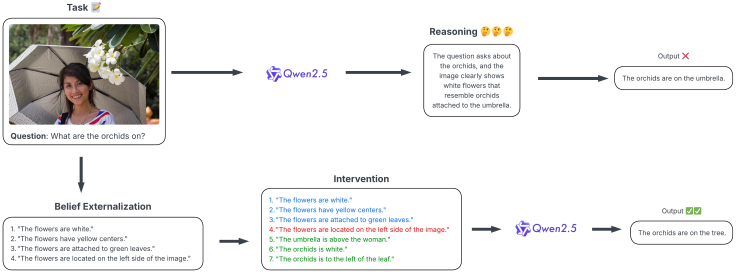
$\hat{A} \neq A^*$ **and** no extracted belief is verifiably false.

Root cause: faulty inference from beliefs

BCI aims to:

1. Quantify the premise-error (PE) rate across VLM failures
2. Provide inference-time interventions that convert PE cases to correct answers

The BCI Pipeline



Five Pipeline Stages

S1 **Belief Ext.**; emit 5–10 visual claims
S2 **Claim Ext.**; typed predicates (e, t, v)
S3 **Verification** vs. scene graph
S4 **Intervention Policy** (E1 / E3 / E12b)
S5 **Re-reasoning**: $\hat{A}' = \mathcal{M}(I, Q, \mathcal{B}')$

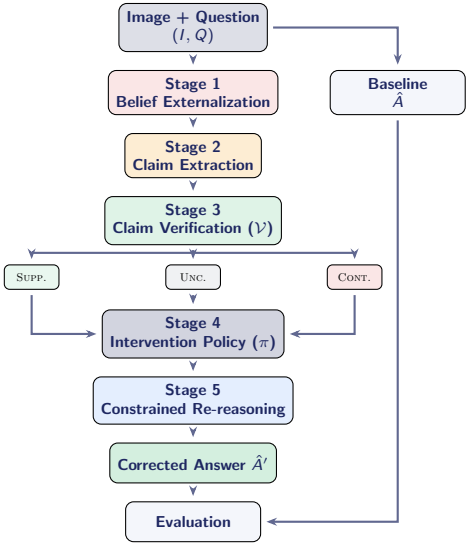
Verifier Labels

- SUPPORTED; matches evidence
- CONTRADICTED; conflicts graph
- UNCERTAIN; insufficient info

Evaluation

Paired statistical evaluation:
McNemar test, ΔAcc, F↑/F↓ on $(\hat{A}, \hat{A}')$ pairs.

BCI: The 5-Stage Pipeline



E1; Replace All

Oracle upper bound: all contradicted claims replaced with GT facts

E3; Remove Only

Conservative: remove contradicted claims without replacement

E12b; Confidence-Gated

Practical: replace only when score $s \geq \tau = 0.18$

Stages 1–3: Externalization, Extraction & Verification

S1: Belief Externalization

$\mathcal{M}$ emits atomic, grounded, falsifiable sentences:

$$\mathcal{B} = \{b_1, \ldots, b_k\}, \quad k \in [5, 10]$$

Examples: "the cup is red", "person is left of the table", "there are three dogs"

S2: Claim Extraction

Beliefs parsed into typed claims (e, t, v):

"cup exists"	`object`
"cup is red"	`attribute`
"cup left of plate"	`relation`
"3 dogs"	`count`

S3: Claim Verification

$$\ell_i = \mathcal{V}(b_i, \mathcal{G}_I) \in \{\textsc{Supp.}, \textsc{Cont.}, \textsc{Unc.}\}$$

Three verifier profiles:

Profile	θ_{attr}	δ_{spatial}
`strict`	0.9	0.10
`balanced`	0.8	0.15
`high_recall`	0.7	0.20

Main experiments use `balanced`.

Design Choice

Verification is **deterministic**; GQA scene graphs as a structured oracle. Cleanly separates verifier design from the intervention hypothesis.

Stage 4: Intervention Policy Family

E1: Replace All
(Causal Upper Bound)

All CONTRADICTED claims replaced with **ground-truth facts** from scene graph.

Oracle policy: requires GT at test time.

+18.0 pp at n=500
+16.2 pp at n=2000

E3: Remove Only
(Conservative Ablation)

CONTRADICTED claims are **deleted** without replacement.

Tests whether removal alone suffices.

−0.6 pp at n=500
(p=0.84, not significant)

E12b: Conf.-Gated
(Practical Variant)

Score $s \in [0, 1]$ from verifier confidence + claim-type reliability.

Replace when $s \geq \tau$ (τ=0.18).

+3.45 pp at n=2000
(p=1.3×10^{-5})

Key Insight

E3 near-zero gain establishes that replacement of false beliefs with correct facts is the operative mechanism; not mere deletion.

Experimental Setup

Dataset: GQA

- Compositional visual questions with structured scene-graph annotations
- 5 question types: `query`, `verify`, `choose`, `logical`, `compare`
- n=500 (main) and n=2000 (large-scale), seed 42

Model: Qwen2.5-VL-7B-Instruct

Model	Base Acc.	Beliefs/Q
LLaVA-1.5-7B	52.2%	2.17
Qwen2.5-VL-7B	**59.2%**	**4.96**

Richer belief externalization enables higher PE detection.

Inference Configuration

- Temperature: 0.0 | Max tokens: 768
- GPU: NVIDIA RTX 6000 Ada (49 GB)
- BCI adds 2 extra VLM passes per sample

Baselines (matched conditions)

- Vanilla (direct VLM)
- Self-Correct
- CoVe-Lite (chain-of-verification)
- Resample-Lite (majority vote, k=3)
- BCI-E3, BCI-E1, BCI-E12b

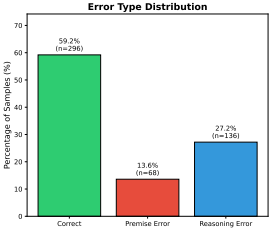
Error Decomposition: How Often Are VLMs Epistemically Wrong?

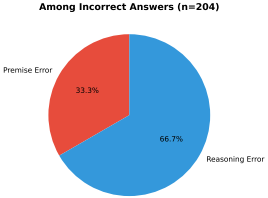
Overall (Qwen2.5-VL, GQA, n=500)

Of 204 incorrect answers:

- **68 (33.3%)** ⇒ **Premise Errors**
- **136 (66.7%)** ⇒ Reasoning Errors

Mean belief coverage: **0.49**
⇒ 33.3% is a **conservative lower bound**





Takeaway

Logical/relational questions most susceptible to premise errors (up to **50%**); exactly the questions depending on precise spatial scene understanding.

Main Intervention Results

Method	n	Acc.	ΔAcc.	F↑	F↓
Vanilla	500	59.2%	—	—	—
Self-Correct	500	49.2%	**−10.0 pp**	—	—
CoVe-Lite	500	57.2%	**−2.0 pp**	—	—
Resample-Lite	500	60.6%	**+1.4 pp**	—	—
BCI-E3	500	58.6%	−0.6 pp	9.4%	10.0%
BCI-E1	500	**77.2%**	**+18.0 pp**	**23.4%**	5.4%
Vanilla	2000	63.55%	—	—	—
BCI-E12b	2000	66.35%	**+3.45 pp**	7.9%	—
BCI-E1	2000	**79.75%**	**+16.2 pp**	**21.2%**	—

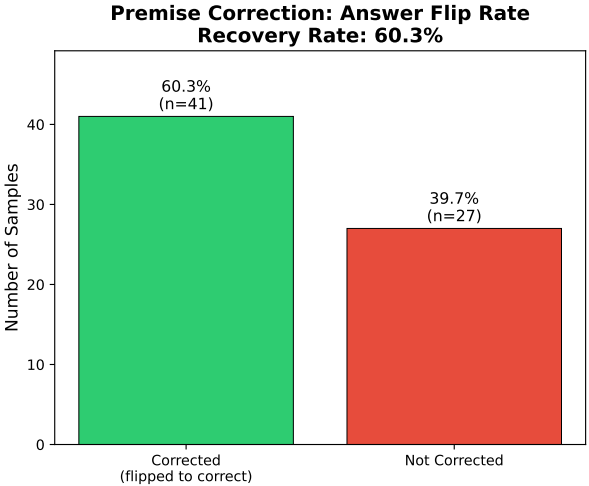
BCI-E1 at n=500

59.2% → 77.2%; 117 flipped correct, 27 regressed.
McNemar $p = 1.5\times10^{-14}$

BCI-E1 at n=2000

63.55% → 79.75%; flip-to-correct **21.2%**
McNemar $p = 1.78\times10^{-48}$

Premise Correction in Action



Recovery Rate

Among the 68 premise-error cases (n=500):

- E1 recovers the correct answer in **60.3%** of PE cases
- Net accuracy: 59.2% → 77.2%
- F↑= 23.4%, F↓= 5.4%

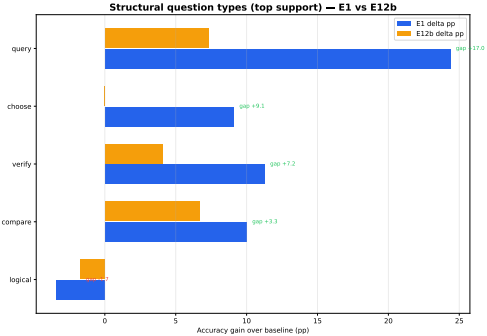
Key Point

Not just overall accuracy; BCI specifically targets and corrects the identifiable subset of *epistemic* failures.

Question-Type Stratification (GQA, n=2000)

E1 vs. E12b by Question Type

Type	n	E1 Δ	E12b Δ
query	887	**+25.9 pp**	+6.2 pp
verify	854	+5.6 pp	+0.4 pp
logical	57	+8.8 pp	+7.0 pp
other	202	**+20.3 pp**	+3.5 pp



Priority A Task 4 — Stratified Analysis

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Task 4 Report: Error-type Stratified Gains (E1 vs E12b)

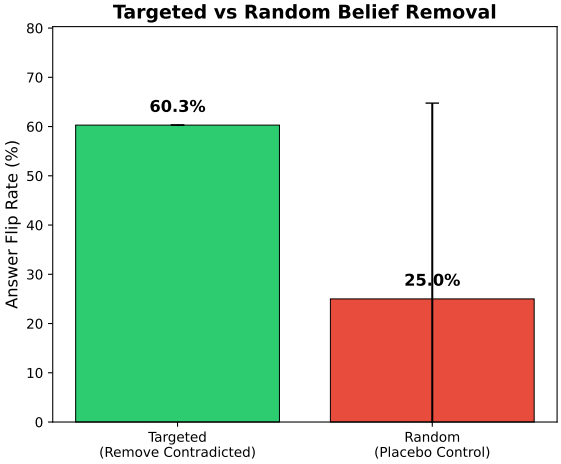
Top structural-type advantages (delta pp gap, E1 - E12b):
- query: +17.03 pp
- choose: +9.09 pp
- verify: +7.21 pp
- compare: +3.33 pp
- logical: -1.72 pp

Top claim-type presence advantages (delta pp gap, E1 - E12b):
- counting: +28.57 pp
- attribute: +18.42 pp
- spatial: +18.42 pp
- object_existence: +16.28 pp
- action: +0.00 pp
```

Semantic Sub-types

Largest E1 advantage: **Categorical** (+20.97 ppgap) and **Relational** (+17.30 ppgap) over E12b.

Targeted vs. Random Correction: Causal Validation



Random Ablation Control

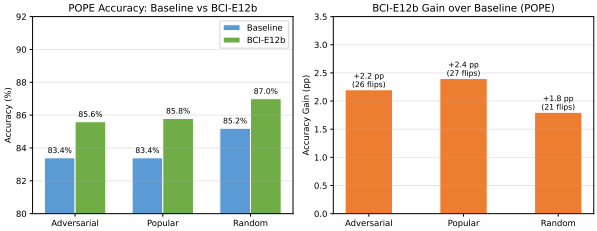
Same removal budget; beliefs deleted at random:

Method	Flip Rate
Random deletion	25.0%
Targeted (E1)	**60.3%**
Gap	**+35.3 pp**

Causal Confirmation

The +35 ppgap **rules out** the hypothesis that *any* belief modification improves answers. The mechanism is grounded in accurate identification and replacement of **false premises**.

Cross-Benchmark Generalization: POPE



POPE (3 splits $\times$ n=500)

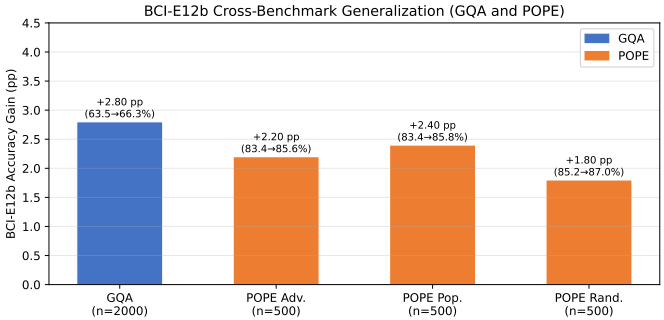
Binary yes/no object-existence questions:

Split	Base	BCI	Δ
Adversarial	83.4%	85.6%	+2.2 pp
Popular	83.4%	85.8%	+2.4 pp
Random	85.2%	87.0%	+1.8 pp

Cross-Benchmark Pattern

All gains positive despite high baseline (>83%).
Premise-error correction is **not a GQA-specific artifact**.

POPE Cross-Benchmark Context



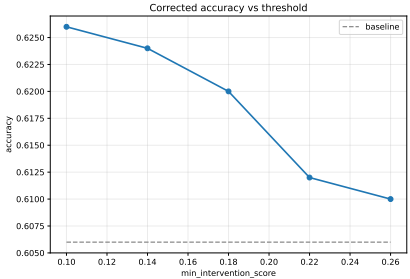
Consistent Pattern

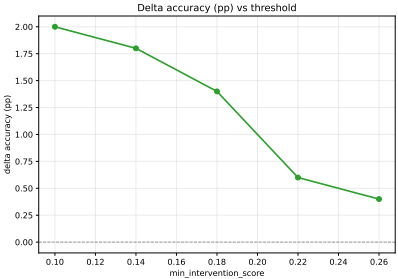
BCI-E12b delivers consistent improvements across both GQA (compositional reasoning) and POPE (object existence), confirming generalizability across structurally distinct benchmarks.

Near-Ceiling Regime

BCI-E12b improves results even when baseline accuracy exceeds 83%; a challenging near-ceiling regime.

Confidence-Gated Policy (E12b): Threshold Robustness





Threshold Selection: τ=0.18

Paired E12 (τ=0.25) vs. E12b (τ=0.18):
eftmargin=*Rate: 35.2% $\rightarrow$ 58.8%eftmargin=*+25 wins, -11 losseseftmargin=*Net: $+14$ samples correctedeftmargin=*McNemar p=0.029

Selectivity

E12b's gating concentrates corrections in $s \in [0.15, 0.30]$ where **correct corrections dominate**.

At n=2000

Multi-seed: 66.18% $\pm$ 0.14pp (seeds 42, 123, 456) — highly reproducible.

Cost-Efficiency Analysis

Method	n	Runs	Seeds	Baseline Acc (mean)	Corrected Acc (mean)	Delta pp (mean)	Runtime per sample	Samples per hour	Flips per hour	Gain pp/hour
E1	0	2						0.00	0.00	
E3	0	1						0.00	0.00	
E12	20	1		45.00%	50.00%	5.000	14.066s	255.94	38.39	63.99
E1	500	4		59.20%	72.55%	13.350	9.773s	371.73	73.65	9.84
E12b	500	1		59.20%	62.40%	3.200	8.714s	413.14	40.49	2.64
E12	500	1		59.20%	59.60%	0.400	7.965s	451.99	22.60	0.36
E3	500	2		59.20%	58.60%	-0.600	8.855s	418.18	39.31	-0.50
E1	2000	3	123,456	63.55%	79.75%	16.200	8.499s	425.00	90.10	3.44
E12b	2000	3	123,456	63.33%	66.18%	2.850	8.525s	424.27	33.38	0.61

Per-Sample Cost (n=2000)

Method	s/sample	Δpp
E1	7.81	+16.2
E12b	7.99	+3.45

E1 delivers 5.8× higher gain at nearly identical latency.

E1 Pareto Dominates

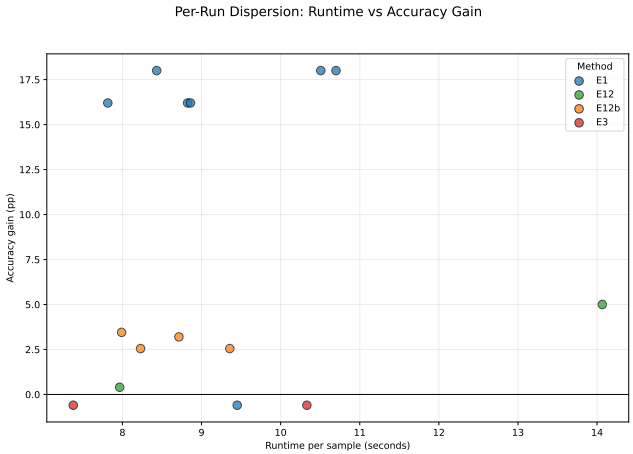
2.07 pp/s vs. 0.43 pp/s for E12b.

90 flips/GPU-hour vs. 33 for E12b.

E12b Use Case

Preferred when Ground Truth facts are unavailable; still delivers significant, reliable gains.

Per-Run Dispersion Across All 18 Runs



Runtime vs. Accuracy Gain

- **E1**: clusters at +16–18 pp, 7.8–10.7 s/sample
- **E12b**: clusters at +2.5–3.5 pp
- **E3**: scatters near zero or negative
- **E12**: hovers near zero

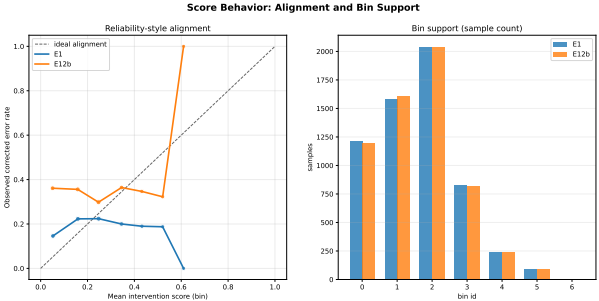
Conclusion

Gain is driven by belief **replacement**, not removal. E3 and E12 are not viable deployment choices.

Reproducibility

All runs logged with git hash, code hash, config hash, and random seed.

Confidence Calibration



Calibration (Reliability Diagram)

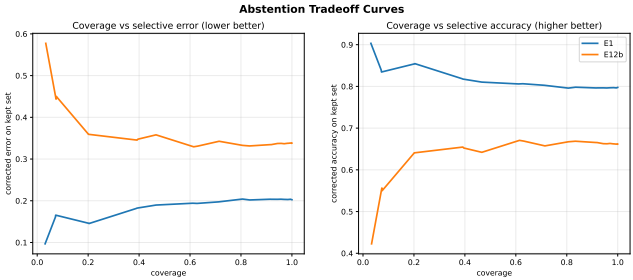
Method	ECE
E1	**0.0793**
E12b	0.1413

E1 closely tracks the ideal calibration diagonal. E12b deviates at high scores due to conditional gating.

Score Distribution

Most samples fall in $s \in [0.1, 0.3]$; a well-calibrated mid-confidence regime.

Abstention Trade-off



Selective Accuracy

Score-based abstention thresholds:

- At **60% coverage**: E1 achieves 80.6% selective accuracy (+16.0 pp)
- At **90% coverage**: E1 maintains 79.6% (+15.8 pp)
- E12b selective: 65.7%–67.1%

Stable Pareto Dominance

E1 maintains ∼+16 ppgain across all coverage levels above 40%. E12b cannot close the gap even selectively.

Statistical Significance: McNemar's Exact Test

Comparison	n	p-value	Significant?
Baseline vs. BCI-E1	500	1.49×10^{-14}	✓
Baseline vs. BCI-E3	500	0.839	✗ (negative control)
Baseline vs. BCI-E1	2000	1.78×10^{-48}	✓
Baseline vs. E12b	2000	1.34×10^{-5}	✓
BCI-E1 vs. E12b	2000	9.99×10^{-43}	✓

E1 and E12b

Both highly significant at n=2000, with E1 reaching $p < 10^{-47}$.

E3: Clean Negative Control

E3 (removal only) is not significant (p=0.84); confirming that **replacement** is the mechanism.

Limitations & Future Work

Current Limitations

- Results limited to **GQA** with deterministic scene-graph verifiers
- **E1 is an oracle**; requires ground-truth facts at test time
- Verifier coverage: 49% mean (low on action & OCR claims)
- Cross-benchmark eval (MMMU, MathVista) remains future work

Future Directions

1. **Learned/Retrieval Verifier**; remove GT dependency
2. **Open-Ended Benchmarks**; MMMU, MathVista, MM-Vet
3. **Uncertainty-Aware Revision**; calibrated belief policies
4. **Cross-Architecture**; InstructBLIP, LLaVA-Next
5. **Pareto Frontier**; full latency/accuracy analysis

Key Takeaways

Empirical Findings

- **33.3%** of wrong VLM answers involve false premises (conservative lower bound)
- Rises to **50%** for logical/relational questions
- E1: **+18.0 pp** (n=500), **+16.2 pp** (n=2000)
- E12b: **+3.45 pp** with 60% intervention coverage
- Targeted > random by **+35 pp**

Methodological Lessons

- Premise errors are a distinct, **actionable** failure mode
- Belief *replacement*; not removal; is the operative mechanism
- Inference-time correction can match training-based fixes
- Structured verifiers enable causal attribution

BCI provides a principled, interpretable, and causally validated pathway to more reliable VLMs; without updating a single model weight.

Thank You

Belief-Constrained Inference (BCI)

Correcting Visual Premises in VLMs at Inference Time

github.com/ksananth4424/BCI_Inference_Time

One-Line Summary

BCI externalizes, verifies, and corrects intermediate visual beliefs in VLMs, delivering up to $+\mathbf{18}$ **pp** accuracy gains without any weight updates, confirmed by rigorous causal ablation.

Appendix: Notation Reference

Symbols

(I, Q, A^*)	image, question, ground truth
$\mathcal{M}$	vision-language model
$\hat{A}$	baseline model answer
$\hat{A}'$	corrected answer
$\mathcal{B}$	set of visual beliefs
b_i	atomic belief
ℓ_i	verifier label for b_i
$\mathcal{G}_I$	scene graph for image I
π	intervention policy
s	intervention score
τ	intervention threshold
ΔAcc	net accuracy change
n	evaluation sample size
F$\uparrow$	flip-to-correct rate
F$\downarrow$	flip-to-wrong rate
θ_{attr}	attribute similarity threshold
δ_{spatial}	spatial tolerance parameter
ECE	expected calibration error

Claim Types

`object`	entity existence
`attribute`	color, shape, material
`relation`	spatial / semantic relations
`action`	activities / verbs
`count`	numerical quantity
`ocr`	text in image

Policies

E1	Replace all contradicted
E3	Remove only (no replace)
E12	Conf.-gated, τ=0.25
E12b	Confidence-gated replace

Verifier Labels

SUPPORTED	claim matches scene graph
CONTRADICTED	claim conflicts with scene graph
UNCERTAIN	insufficient evidence

Appendix: Full Significance Table

Comparison	n	p-**value**	**Notes**
Baseline vs. BCI-E1	500	1.49×10^{-14}	Highly significant
Baseline vs. BCI-E3	500	0.839	Negative control
Baseline vs. BCI-E1	2000	1.78×10^{-48}	Scales robustly
Baseline vs. E12b	2000	1.34×10^{-5}	Holm-corrected
BCI-E1 vs. E12b	2000	9.99×10^{-43}	Different operating pts
E12 vs. E12b (τ)	500	0.0288	Threshold justification